

## Relative Kinematics Computations Model

5.4

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# Chapter 1

## Module Index

### 1.1 Modules

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## Chapter 2

# Namespace Index

### 2.1 Namespace List

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## Chapter 3

# Data Structure Index

### 3.1 Data Structures

Here are the data structures with brief descriptions:

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## Chapter 4

# File Index

### 4.1 File List

Here is a list of all files with brief descriptions:

|                                        |                                                                                                                                                                                                 |    |
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| <a href="#">rel_kin_messages.hh</a>    | Define the class RelKinMessages, the class that specifies the message IDs used in the relative kinematics model . . . . .                                                                       | 22 |
| <a href="#">relative_kinematics.cc</a> | Define methods for the RelativeKinematics class . . . . .                                                                                                                                       | 22 |
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## Chapter 5

# Module Documentation

### 5.1 Models

#### Modules

- [Dynamics](#)

#### 5.1.1 Detailed Description

### 5.2 Dynamics

#### Modules

- [RelKin](#)

#### 5.2.1 Detailed Description

### 5.3 RelKin

#### Files

- file [rel\\_kin\\_messages.hh](#)  
*Define the class RelKinMessages, the class that specifies the message IDs used in the relative kinematics model.*
- file [relative\\_kinematics.hh](#)  
*Define the class RelativeKinematics, the class used for calculating the state of some point(s) of interest associated with the subject DynBody relative to some other reference frame.*
- file [rel\\_kin\\_messages.cc](#)  
*Implement the class RelKinMessages.*
- file [relative\\_kinematics.cc](#)  
*Define methods for the RelativeKinematics class.*

#### 5.3.1 Detailed Description



## Chapter 6

# Namespace Documentation

### 6.1 jeod Namespace Reference

Namespace jeod.

#### Data Structures

- class [RelKinMessages](#)  
*Specifies the message IDs used in the orbital elements model.*
- class [RelativeKinematics](#)  
*Encapsulates functionality for computing relative states.*

#### 6.1.1 Detailed Description

Namespace jeod.





## Chapter 7

# Data Structure Documentation

### 7.1 jeod::RelativeKinematics Class Reference

Encapsulates functionality for computing relative states.

```
#include <relative_kinematics.hh>
```

#### Public Member Functions

- [RelativeKinematics](#) ()  
*Construct a [RelativeKinematics](#) object.*
- [~RelativeKinematics](#) ()  
*Destruct a [RelativeKinematics](#) object.*
- [RelativeKinematics](#) (const [RelativeKinematics](#) &)=delete
- [RelativeKinematics](#) & operator= (const [RelativeKinematics](#) &)=delete
- void [add\\_relstate](#) (RelativeDerivedState &relstate)  
*Add a relative state to the list of ones maintained by this model.*
- void [remove\\_relstate](#) (RelativeDerivedState &relstate)  
*Remove a relative state from the list of ones maintained by this model.*
- RelativeDerivedState \* [find\\_relstate](#) (const std::string &relstate\_name)  
*Find a specific relative state maintained by this model.*
- void [activate\\_relstate](#) (RelativeDerivedState &relstate, bool raf)  
*Set flag for a relative state to be activated or deactivated by the RelKin manager.*
- void [update\\_single](#) (const std::string &relstate\_name)  
*Update a single relative state maintained by this model.*
- void [update\\_all](#) ()  
*Update all relative states maintained by this model.*

#### Data Fields

- unsigned int [num\\_rel\\_states](#) {}  
*Length of above list of relative states being maintained by this.*
- JeodPointerVector< RelativeDerivedState >::type [relative\\_states](#)  
*List of relative states to be computed and maintained by this model.*

## Private Member Functions

- [JEOD\\_CLASS\\_ESTABLISH\\_FRIENDS2](#) (jeod, [RelativeKinematics](#))

### 7.1.1 Detailed Description

Encapsulates functionality for computing relative states.

Definition at line 85 of file `relative_kinematics.hh`.

### 7.1.2 Constructor & Destructor Documentation

#### 7.1.2.1 [RelativeKinematics\(\)](#) [1/2]

```
jeod::RelativeKinematics::RelativeKinematics ( )
```

Construct a [RelativeKinematics](#) object.

Definition at line 55 of file `relative_kinematics.cc`.

References `relative_states`.

#### 7.1.2.2 [~RelativeKinematics\(\)](#)

```
jeod::RelativeKinematics::~~RelativeKinematics ( )
```

Destruct a [RelativeKinematics](#) object.

Definition at line 65 of file `relative_kinematics.cc`.

References `relative_states`.

#### 7.1.2.3 [RelativeKinematics\(\)](#) [2/2]

```
jeod::RelativeKinematics::RelativeKinematics (
    const RelativeKinematics & ) [delete]
```

### 7.1.3 Member Function Documentation

#### 7.1.3.1 [activate\\_relstate\(\)](#)

```
void jeod::RelativeKinematics::activate_relstate (
    RelativeDerivedState & relstate,
    bool raf )
```

Set flag for a relative state to be activated or deactivated by the RelKin manager.

**Parameters**

|    |                 |                                 |
|----|-----------------|---------------------------------|
| in | <i>relstate</i> | Relstate to activate/deactivate |
| in | <i>raf</i>      | bool Relstate activation flag   |

Definition at line 151 of file `relative_kinematics.cc`.

References `jeod::RelKinMessages::entry_not_found`, `find_relstate()`, and `jeod::RelKinMessages::invalid_entry`.

**7.1.3.2 add\_relstate()**

```
void jeod::RelativeKinematics::add_relstate (
    RelativeDerivedState & relstate )
```

Add a relative state to the list of ones maintained by this model.

**Parameters**

|    |                 |                 |
|----|-----------------|-----------------|
| in | <i>relstate</i> | Relstate to add |
|----|-----------------|-----------------|

Definition at line 74 of file `relative_kinematics.cc`.

References `jeod::RelKinMessages::duplicate_entry`, `find_relstate()`, `num_rel_states`, and `relative_states`.

**7.1.3.3 find\_relstate()**

```
RelativeDerivedState * jeod::RelativeKinematics::find_relstate (
    const std::string & relstate_name )
```

Find a specific relative state maintained by this model.

**Returns**

Void

**Parameters**

|    |                      |                  |
|----|----------------------|------------------|
| in | <i>relstate_name</i> | Relstate to find |
|----|----------------------|------------------|

Definition at line 128 of file `relative_kinematics.cc`.

References `num_rel_states`, and `relative_states`.

Referenced by `activate_relstate()`, `add_relstate()`, and `update_single()`.

#### 7.1.3.4 JEOD\_CLASS\_ESTABLISH\_FRIENDS2()

```
jeod::RelativeKinematics::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    RelativeKinematics ) [private]
```

#### 7.1.3.5 operator=()

```
RelativeKinematics& jeod::RelativeKinematics::operator= (
    const RelativeKinematics & ) [delete]
```

#### 7.1.3.6 remove\_relstate()

```
void jeod::RelativeKinematics::remove_relstate (
    RelativeDerivedState & relstate )
```

Remove a relative state from the list of ones maintained by this model.

##### Parameters

|    |                 |                    |
|----|-----------------|--------------------|
| in | <i>relstate</i> | Relstate to remove |
|----|-----------------|--------------------|

Definition at line 105 of file `relative_kinematics.cc`.

References `jeod::RelKinMessages::entry_not_found`, `num_rel_states`, and `relative_states`.

#### 7.1.3.7 update\_all()

```
void jeod::RelativeKinematics::update_all ( )
```

Update all relative states maintained by this model.

relstates that have been deactivated from RelKin will not be update.

Definition at line 198 of file `relative_kinematics.cc`.

References `num_rel_states`, and `relative_states`.

#### 7.1.3.8 update\_single()

```
void jeod::RelativeKinematics::update_single (
    const std::string & relstate_name )
```

Update a single relative state maintained by this model.

## Parameters

|    |                      |                    |
|----|----------------------|--------------------|
| in | <i>relstate_name</i> | Relstate to update |
|----|----------------------|--------------------|

Definition at line 187 of file `relative_kinematics.cc`.

References `find_relstate()`.

## 7.1.4 Field Documentation

### 7.1.4.1 num\_rel\_states

```
unsigned int jeod::RelativeKinematics::num_rel_states {}
```

Length of above list of relative states being maintained by this.

`trick_units(-)`

Definition at line 92 of file `relative_kinematics.hh`.

Referenced by `add_relstate()`, `find_relstate()`, `remove_relstate()`, and `update_all()`.

### 7.1.4.2 relative\_states

```
JeodPointerVector<RelativeDerivedState>::type jeod::RelativeKinematics::relative_states
```

List of relative states to be computed and maintained by this model.

Note that this list is not restricted to be relative states associated with only a single `DynBody`. `trick_io(**)`

Definition at line 99 of file `relative_kinematics.hh`.

Referenced by `add_relstate()`, `find_relstate()`, `RelativeKinematics()`, `remove_relstate()`, `update_all()`, and `~RelativeKinematics()`.

The documentation for this class was generated from the following files:

- [relative\\_kinematics.hh](#)
- [relative\\_kinematics.cc](#)

## 7.2 jeod::RelKinMessages Class Reference

Specifies the message IDs used in the orbital elements model.

```
#include <rel_kin_messages.hh>
```

## Public Member Functions

- [RelKinMessages](#) ()=delete
- [RelKinMessages](#) (const [RelKinMessages](#) &)=delete
- [RelKinMessages](#) & operator= (const [RelKinMessages](#) &)=delete

## Static Public Attributes

- static const char \* [duplicate\\_entry](#) = "dynamics/rel\_kin/" "duplicate\_entry"  
*Issued when a duplicate entry is found.*
- static const char \* [entry\\_not\\_found](#) = "dynamics/rel\_kin/" "entry\_not\_found"  
*Issued when an entry is not found.*
- static const char \* [invalid\\_entry](#) = "dynamics/rel\_kin/" "invalid\_entry"  
*Issued when function input is invalid.*

## Private Member Functions

- [JEOD\\_CLASS\\_ESTABLISH\\_FRIENDS2](#) (jeod, [RelKinMessages](#))

### 7.2.1 Detailed Description

Specifies the message IDs used in the orbital elements model.

Definition at line 81 of file `rel_kin_messages.hh`.

### 7.2.2 Constructor & Destructor Documentation

#### 7.2.2.1 RelKinMessages() [1/2]

```
jeod::RelKinMessages::RelKinMessages ( ) [delete]
```

#### 7.2.2.2 RelKinMessages() [2/2]

```
jeod::RelKinMessages::RelKinMessages (
    const RelKinMessages & ) [delete]
```

### 7.2.3 Member Function Documentation

### 7.2.3.1 JEOD\_CLASS\_ESTABLISH\_FRIENDS2()

```
jeod::RelKinMessages::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    RelKinMessages ) [private]
```

### 7.2.3.2 operator=()

```
RelKinMessages& jeod::RelKinMessages::operator= (
    const RelKinMessages & ) [delete]
```

## 7.2.4 Field Documentation

### 7.2.4.1 duplicate\_entry

```
char const * jeod::RelKinMessages::duplicate_entry = "dynamics/rel_kin/" "duplicate_entry"
[static]
```

Issued when a duplicate entry is found.

trick\_units(–)

Definition at line 88 of file rel\_kin\_messages.hh.

Referenced by jeod::RelativeKinematics::add\_relstate().

### 7.2.4.2 entry\_not\_found

```
char const * jeod::RelKinMessages::entry_not_found = "dynamics/rel_kin/" "entry_not_found"
[static]
```

Issued when an entry is not found.

trick\_units(–)

Definition at line 93 of file rel\_kin\_messages.hh.

Referenced by jeod::RelativeKinematics::activate\_relstate(), and jeod::RelativeKinematics::remove\_relstate().

### 7.2.4.3 invalid\_entry

```
char const * jeod::RelKinMessages::invalid_entry = "dynamics/rel_kin/" "invalid_entry" [static]
```

Issued when function input is invalid.

trick\_units(–)

Definition at line 98 of file rel\_kin\_messages.hh.

Referenced by jeod::RelativeKinematics::activate\_relstate().

The documentation for this class was generated from the following files:

- [rel\\_kin\\_messages.hh](#)
- [rel\\_kin\\_messages.cc](#)





## Chapter 8

# File Documentation

### 8.1 rel\_kin\_messages.cc File Reference

Implement the class RelKinMessages.

```
#include "utils/message/include/make_message_code.hh"
#include "../include/rel_kin_messages.hh"
```

#### Namespaces

- [jeod](#)  
*Namespace jeod.*

#### Macros

- `#define`     [MAKE\\_RELKIN\\_MESSAGE\\_CODE](#)(id)     JEOD\_MAKE\_MESSAGE\_CODE(RelKinMessages, "dynamics/rel\_kin/", id)

#### 8.1.1 Detailed Description

Implement the class RelKinMessages.

#### 8.1.2 Macro Definition Documentation

##### 8.1.2.1 MAKE\_RELKIN\_MESSAGE\_CODE

```
#define MAKE_RELKIN_MESSAGE_CODE(  
    id ) JEOD_MAKE_MESSAGE_CODE(RelKinMessages, "dynamics/rel_kin/", id)
```

Definition at line 43 of file rel\_kin\_messages.cc.

## 8.2 rel\_kin\_messages.hh File Reference

Define the class RelKinMessages, the class that specifies the message IDs used in the relative kinematics model.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

### Data Structures

- class [jeod::RelKinMessages](#)  
*Specifies the message IDs used in the orbital elements model.*

### Namespaces

- [jeod](#)  
*Namespace jeod.*

#### 8.2.1 Detailed Description

Define the class RelKinMessages, the class that specifies the message IDs used in the relative kinematics model.

## 8.3 relative\_kinematics.cc File Reference

Define methods for the RelativeKinematics class.

```
#include <algorithm>
#include <cstdint>
#include "dynamics/derived_state/include/relative_derived_state.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/rel_kin_messages.hh"
#include "../include/relative_kinematics.hh"
```

### Namespaces

- [jeod](#)  
*Namespace jeod.*

#### 8.3.1 Detailed Description

Define methods for the RelativeKinematics class.

## 8.4 relative\_kinematics.hh File Reference

Define the class RelativeKinematics, the class used for calculating the state of some point(s) of interest associated with the subject DynBody relative to some other reference frame.

```
#include "dynamics/derived_state/include/class_declarations.hh"
#include "utils/container/include/pointer_vector.hh"
#include "utils/sim_interface/include/jeod_class.hh"
```

### Data Structures

- class [jeod::RelativeKinematics](#)  
*Encapsulates functionality for computing relative states.*

### Namespaces

- [jeod](#)  
*Namespace jeod.*

#### 8.4.1 Detailed Description

Define the class RelativeKinematics, the class used for calculating the state of some point(s) of interest associated with the subject DynBody relative to some other reference frame.



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